

Function / Loop / DateTime

Question:

Write a C# console application that defines a **function** to display the calendar for a given month and year. The function should accept two parameters: **month** and **year**, and display the calendar in a table format. The table should include the days of the week (Sunday, Monday, etc.) as headers, followed by the days of the month arranged under the appropriate weekday. call this function in Main() using proper input. (using **nested loop** and **break**)

If the input is:

```
Enter month (1-12): 1
Enter year: 2025
```

The output should be:

```
Calendar for 1/2025
```

```
Sun Mon Tue Wed Thu Fri Sat
                    1 2 3 4
    5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

Array

Question:

Write a C# program that uses a **jagged array** to track the student attendance for each day. For each day, the program should accept the number of students absent and then store the roll numbers of the students who were absent. The program should display the absent roll numbers for each day.

The program should follow these steps:

1. Accept the number of days from the user.
2. For each day, accept the number of students who were absent.
3. Accept the roll numbers of the absent students for that day.
4. Store the absent roll numbers for each day in a jagged array.
5. Display the absent roll numbers for each day.

Example Input and Output:

Input:

Enter number of days: 3

Enter attendance for Day 1:

Enter number of students absent: 2

Enter absent roll numbers: 1 3

Enter attendance for Day 2:

Enter number of students absent: 1

Enter absent roll numbers: 2

Enter attendance for Day 3:

Enter number of students absent: 3

Enter absent roll numbers: 1 2 3

Output:

Absent roll numbers for Day 1: 1 3

Absent roll numbers for Day 2: 2

Absent roll numbers for Day 3: 1 2 3

Function

Question:

Write a C# program that demonstrates the usage of **default** and **ref** parameters in functions.

The program should have two functions:

1. **Function with a default parameter:** The function should accept a student's name and an optional grade. If no grade is provided, it should default to "B".
2. **Function with a ref parameter:** The function should accept a student's score and update the score by adding a bonus based on the grade. The grade will be used to determine the bonus: if the grade is "A", add 10 points; if it's "B", add 5 points; if it's "C", no bonus will be added.

The program should:

- Accept the student's name and grade (optional) as inputs.
- Use the function with the default parameter to display the student's name and grade. •

Use the function with the ref parameter to calculate the new score based on the grade.

Example Input and Output:

Input:

```
Enter student's name: John
Enter grade (optional, default is B): A
Enter initial score: 80
```

Output:

```
Student: John, Grade: A
Initial Score: 80
Updated Score with Bonus: 90
```

Here's a question that involves **this**, **constructors**, **properties**, and **static members** in C# within a class:

Question:

Write a C# program that defines a class **Student** to manage student information. The class should use the following features:

1. **this keyword**: Use the **this** keyword to refer to instance variables and differentiate between instance variables and parameters.
2. **Constructor**: Create a constructor to initialize the student's name, age, and grade.
3. **Properties**: Implement properties for the student's name, age, and grade, with validation for the grade (it should only be A, B, or C, if not valid assign "").
4. **Static Member**: Implement a static member to keep track of the total number of students created.